Informatica

Informatica® PowerExchange for Tableau
10.5.10

User Guide for PowerCenter

Informatica PowerExchange for Tableau User Guide for PowerCenter
10.5.10
February 2026

© Copyright Informatica LLC 2015, 2025

This software and documentation are provided only under a separate license agreement containing restrictions on use and disclosure. No part of this document may be reproduced or transmitted in any form, by any means (electronic, photocopying, recording or otherwise) without prior consent of Informatica LLC.

Informatica, the Informatica logo, PowerCenter, and PowerExchange are trademarks or registered trademarks of Informatica LLC in the United States and many jurisdictions throughout the world. A current list of Informatica trademarks is available on the web at https://www.informatica.com/trademarks.html. Other company and product names may be trade names or trademarks of their respective owners.

U.S. GOVERNMENT RIGHTS Programs, software, databases, and related documentation and technical data delivered to U.S. Government customers are "commercial computer software" or "commercial technical data" pursuant to the applicable Federal Acquisition Regulation and agency-specific supplemental regulations. As such, the use, duplication, disclosure, modification, and adaptation is subject to the restrictions and license terms set forth in the applicable Government contract, and, to the extent applicable by the terms of the Government contract, the additional rights set forth in FAR 52.227-19, Commercial Computer Software License.

Portions of this software and/or documentation are subject to copyright held by third parties, including without limitation: Copyright DataDirect Technologies. All rights reserved. Copyright © Sun Microsystems. All rights reserved. Copyright © RSA Security Inc. All Rights Reserved. Copyright © Ordinal Technology Corp. All rights reserved. Copyright © Aandacht c.v. All rights reserved. Copyright Genivia, Inc. All rights reserved. Copyright Isomorphic Software. All rights reserved. Copyright © Meta Integration Technology, Inc. All rights reserved. Copyright © Intalio. All rights reserved. Copyright © Oracle. All rights reserved. Copyright © Adobe Systems Incorporated. All rights reserved. Copyright © DataArt, Inc. All rights reserved. Copyright © ComponentSource. All rights reserved. Copyright © Microsoft Corporation. All rights reserved. Copyright © Rogue Wave Software, Inc. All rights reserved. Copyright © Teradata Corporation. All rights reserved. Copyright © Yahoo! Inc. All rights reserved. Copyright © Glyph & Cog, LLC. All rights reserved. Copyright © Thinkmap, Inc. All rights reserved. Copyright © Clearpace Software Limited. All rights reserved. Copyright © Information Builders, Inc. All rights reserved. Copyright © OSS Nokalva, Inc. All rights reserved. Copyright Edifecs, Inc. All rights reserved. Copyright Cleo Communications, Inc. All rights reserved. Copyright © International Organization for Standardization 1986. All rights reserved. Copyright © ej-technologies GmbH. All rights reserved. Copyright © Jaspersoft Corporation. All rights reserved. Copyright © International Business Machines Corporation. All rights reserved. Copyright © yWorks GmbH. All rights reserved. Copyright © Lucent Technologies. All rights reserved. Copyright © University of Toronto. All rights reserved. Copyright © Daniel Veillard. All rights reserved. Copyright © Unicode, Inc. Copyright IBM Corp. All rights reserved. Copyright © MicroQuill Software Publishing, Inc. All rights reserved. Copyright © PassMark Software Pty Ltd. All rights reserved. Copyright © LogiXML, Inc. All rights reserved. Copyright © 2003-2010 Lorenzi Davide, All rights reserved. Copyright © Red Hat, Inc. All rights reserved. Copyright © The Board of Trustees of the Leland Stanford Junior University. All rights reserved. Copyright © EMC Corporation. All rights reserved. Copyright © Flexera Software. All rights reserved. Copyright © Jinfonet Software. All rights reserved. Copyright © Apple Inc. All rights reserved. Copyright © Telerik Inc. All rights reserved. Copyright © BEA Systems. All rights reserved. Copyright © PDFlib GmbH. All rights reserved. Copyright © Orientation in Objects GmbH. All rights reserved. Copyright © Tanuki Software, Ltd. All rights reserved. Copyright © Ricebridge. All rights reserved. Copyright © Sencha, Inc. All rights reserved. Copyright © Scalable Systems, Inc. All rights reserved. Copyright © jQWidgets. All rights reserved. Copyright © Tableau Software, Inc. All rights reserved. Copyright© MaxMind, Inc. All Rights Reserved. Copyright © TMate Software s.r.o. All rights reserved. Copyright © MapR Technologies Inc. All rights reserved. Copyright © Amazon Corporate LLC. All rights reserved. Copyright © Highsoft. All rights reserved. Copyright © Python Software Foundation. All rights reserved. Copyright © BeOpen.com. All rights reserved. Copyright © CNRI. All rights reserved.

This product includes software developed by the Apache Software Foundation (http://www.apache.org/), and/or other software which is licensed under various versions of the Apache License (the "License"). You may obtain a copy of these Licenses at http://www.apache.org/licenses/. Unless required by applicable law or agreed to in writing, software distributed under these Licenses is distributed on an "AS IS" BASIS, WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied. See the Licenses for the specific language governing permissions and limitations under the Licenses.

This product includes software which was developed by Mozilla (http://www.mozilla.org/), software copyright The JBoss Group, LLC, all rights reserved; software copyright © 1999-2006 by Bruno Lowagie and Paulo Soares and other software which is licensed under various versions of the GNU Lesser General Public License Agreement, which may be found at http:// www.gnu.org/licenses/lgpl.html. The materials are provided free of charge by Informatica, "as-is", without warranty of any kind, either express or implied, including but not limited to the implied warranties of merchantability and fitness for a particular purpose.

The product includes ACE(TM) and TAO(TM) software copyrighted by Douglas C. Schmidt and his research group at Washington University, University of California, Irvine, and Vanderbilt University, Copyright (©) 1993-2006, all rights reserved.

This product includes software developed by the OpenSSL Project for use in the OpenSSL Toolkit (copyright The OpenSSL Project. All Rights Reserved) and redistribution of this software is subject to terms available at http://www.openssl.org and http://www.openssl.org/source/license.html.

This product includes Curl software which is Copyright 1996-2013, Daniel Stenberg, <daniel@haxx.se>. All Rights Reserved. Permissions and limitations regarding this software are subject to terms available at http://curl.haxx.se/docs/copyright.html. Permission to use, copy, modify, and distribute this software for any purpose with or without fee is hereby granted, provided that the above copyright notice and this permission notice appear in all copies.

The product includes software copyright 2001-2005 (©) MetaStuff, Ltd. All Rights Reserved. Permissions and limitations regarding this software are subject to terms available at http://www.dom4j.org/ license.html.

The product includes software copyright © 2004-2007, The Dojo Foundation. All Rights Reserved. Permissions and limitations regarding this software are subject to terms available at http://dojotoolkit.org/license.

This product includes ICU software which is copyright International Business Machines Corporation and others. All rights reserved. Permissions and limitations regarding this software are subject to terms available at http://source.icu-project.org/repos/icu/icu/trunk/license.html.

This product includes software copyright © 1996-2006 Per Bothner. All rights reserved. Your right to use such materials is set forth in the license which may be found at http:// www.gnu.org/software/ kawa/Software-License.html.

This product includes OSSP UUID software which is Copyright © 2002 Ralf S. Engelschall, Copyright © 2002 The OSSP Project Copyright © 2002 Cable & Wireless Deutschland. Permissions and limitations regarding this software are subject to terms available at http://www.opensource.org/licenses/mit-license.php.

This product includes software developed by Boost (http://www.boost.org/) or under the Boost software license. Permissions and limitations regarding this software are subject to terms available at http:/ /www.boost.org/LICENSE_1_0.txt.

This product includes software copyright © 1997-2007 University of Cambridge. Permissions and limitations regarding this software are subject to terms available at http:// www.pcre.org/license.txt.

This product includes software copyright © 2007 The Eclipse Foundation. All Rights Reserved. Permissions and limitations regarding this software are subject to terms available at http:// www.eclipse.org/org/documents/epl-v10.php and at http://www.eclipse.org/org/documents/edl-v10.php.

This product includes software licensed under the terms at http://www.tcl.tk/software/tcltk/license.html, http://www.bosrup.com/web/overlib/?License, http:// www.stlport.org/doc/ license.html, http://asm.ow2.org/license.html, http://www.cryptix.org/LICENSE.TXT, http://hsqldb.org/web/hsqlLicense.html, http:// httpunit.sourceforge.net/doc/ license.html, http://jung.sourceforge.net/license.txt , http://www.gzip.org/zlib/zlib_license.html, http://www.openldap.org/software/ release/license.html, http://www.libssh2.org, http://slf4j.org/license.html, http://www.sente.ch/software/OpenSourceLicense.html, http://fusesource.com/downloads/ license-agreements/fuse-message-broker-v-5-3- license-agreement; http://antlr.org/license.html; http://aopalliance.sourceforge.net/; http://www.bouncycastle.org/ licence.html; http://www.jgraph.com/jgraphdownload.html; http://www.jcraft.com/jsch/LICENSE.txt; http://jotm.objectweb.org/bsd_license.html; . http://www.w3.org/

Consortium/Legal/2002/copyright-software-20021231; http://www.slf4j.org/license.html; http://nanoxml.sourceforge.net/orig/copyright.html; http://www.json.org/license.html; http://forge.ow2.org/projects/javaservice/, http://www.postgresql.org/about/licence.html, http://www.sqlite.org/copyright.html, http://www.tcl.tk/software/tcltk/license.html, http://www.jaxen.org/faq.html, http://www.jdom.org/docs/faq.html, http://www.slf4j.org/license.html; http://www.iodbc.org/dataspace/iodbc/wiki/iODBC/License; http://www.keplerproject.org/md5/license.html; http://www.toedter.com/en/jcalendar/license.html; http://www.edankert.com/bounce/index.html; http://www.net-snmp.org/about/license.html; http://www.openmdx.org/#FAQ; http://www.php.net/license/3_01.txt; http://srp.stanford.edu/license.txt; http://www.schneier.com/blowfish.html; http://www.jmock.org/license.html; http://xsom.java.net; http://benalman.com/about/license/; https://github.com/CreateJS/EaselJS/blob/master/src/easeljs/display/Bitmap.js; http://www.h2database.com/html/license.html#summary; http://jsoncpp.sourceforge.net/LICENSE; http://jdbc.postgresql.org/license.html; http://protobuf.googlecode.com/svn/trunk/src/google/protobuf/descriptor.proto; https://github.com/rantav/hector/blob/master/LICENSE; http://web.mit.edu/Kerberos/krb5-current/doc/mitK5license.html; http://jibx.sourceforge.net/jibx-license.html; https://github.com/lyokato/libgeohash/blob/master/LICENSE; https://github.com/hjiang/jsonxx/blob/master/LICENSE; https://code.google.com/p/lz4/; https://github.com/jedisct1/libsodium/blob/master/LICENSE; http://one-jar.sourceforge.net/index.php?page=documents&file=license; https://github.com/EsotericSoftware/kryo/blob/master/license.txt; http://www.scala-lang.org/license.html; https://github.com/tinkerpop/blueprints/blob/master/LICENSE.txt; http://gee.cs.oswego.edu/dl/classes/EDU/oswego/cs/dl/util/concurrent/intro.html; https://aws.amazon.com/asl/; https://github.com/twbs/bootstrap/blob/master/LICENSE; https://sourceforge.net/p/xmlunit/code/HEAD/tree/trunk/LICENSE.txt; https://github.com/documentcloud/underscore-contrib/blob/master/LICENSE, and https://github.com/apache/hbase/blob/master/LICENSE.txt.

This product includes software licensed under the Academic Free License (http://www.opensource.org/licenses/afl-3.0.php), the Common Development and Distribution License (http://www.opensource.org/licenses/cddl1.php) the Common Public License (http://www.opensource.org/licenses/cpl1.0.php), the Sun Binary Code License Agreement Supplemental License Terms, the BSD License (http:// www.opensource.org/licenses/bsd-license.php), the new BSD License (http://opensource.org/licenses/BSD-3-Clause), the MIT License (http://www.opensource.org/licenses/mit-license.php), the Artistic License (http://www.opensource.org/licenses/artistic-license-1.0) and the Initial Developer's Public License Version 1.0 (http://www.firebirdsql.org/en/initial-developer-s-public-license-version-1-0/).

This product includes software copyright © 2003-2006 Joe Walnes, 2006-2007 XStream Committers. All rights reserved. Permissions and limitations regarding this software are subject to terms available at http://xstream.codehaus.org/license.html. This product includes software developed by the Indiana University Extreme! Lab. For further information please visit http://www.extreme.indiana.edu/.

This product includes software Copyright (c) 2013 Frank Balluffi and Markus Moeller. All rights reserved. Permissions and limitations regarding this software are subject to terms of the MIT license.

See patents at https://www.informatica.com/legal/patents.html.

DISCLAIMER: Informatica LLC provides this documentation "as is" without warranty of any kind, either express or implied, including, but not limited to, the implied warranties of noninfringement, merchantability, or use for a particular purpose. Informatica LLC does not warrant that this software or documentation is error free. The information provided in this software or documentation may include technical inaccuracies or typographical errors. The information in this software and documentation is subject to change at any time without notice.

NOTICES

This Informatica product (the "Software") includes certain drivers (the "DataDirect Drivers") from DataDirect Technologies, an operating company of Progress Software Corporation ("DataDirect") which are subject to the following terms and conditions:

1. THE DATADIRECT DRIVERS ARE PROVIDED "AS IS" WITHOUT WARRANTY OF ANY KIND, EITHER EXPRESSED OR IMPLIED, INCLUDING BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE AND NON-INFRINGEMENT.
2. IN NO EVENT WILL DATADIRECT OR ITS THIRD PARTY SUPPLIERS BE LIABLE TO THE END-USER CUSTOMER FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, CONSEQUENTIAL OR OTHER DAMAGES ARISING OUT OF THE USE OF THE ODBC DRIVERS, WHETHER OR NOT INFORMED OF THE POSSIBILITIES OF DAMAGES IN ADVANCE. THESE LIMITATIONS APPLY TO ALL CAUSES OF ACTION, INCLUDING, WITHOUT LIMITATION, BREACH OF CONTRACT, BREACH OF WARRANTY, NEGLIGENCE, STRICT LIABILITY, MISREPRESENTATION AND OTHER TORTS.

The information in this documentation is subject to change without notice. If you find any problems in this documentation, report them to us at infa_documentation@informatica.com.

Informatica products are warranted according to the terms and conditions of the agreements under which they are provided. INFORMATICA PROVIDES THE INFORMATION IN THIS DOCUMENT "AS IS" WITHOUT WARRANTY OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING WITHOUT ANY WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE AND ANY WARRANTY OR CONDITION OF NON-INFRINGEMENT.

Publication Date: 2026-02-19

Table of Contents

Preface 6

Informatica Resources 6

- Informatica Network 6
- Informatica Knowledge Base 6
- Informatica Documentation 6
- Informatica Product Availability Matrices 7
- Informatica Velocity 7
- Informatica Marketplace 7
- Informatica Global Customer Support 7

Chapter 1: Introduction to PowerExchange for Tableau 8

PowerExchange for Tableau Overview 8

Introduction to Tableau 8

PowerExchange for Tableau Implementation 9

PowerExchange for Tableau Example 9

Chapter 2: PowerExchange for Tableau Configuration 10

PowerExchange for Tableau Configuration Overview 10

PowerExchange for Tableau Plug-in Registration 10

- Registering the Plug-in from the Administrator Tool 10
- Registering the Plug-in from the Command Line Program 11

Upgrading PowerExchange for Tableau 11

Chapter 3: Tableau Targets 12

Tableau Targets Overview 12

Importing Tableau Target Definitions 12

Chapter 4: Tableau Mappings 13

Tableau Mappings Overview 13

Tableau Mapping Example 13

Chapter 5: Tableau Sessions 15

Tableau Session Overview 15

PowerExchange for Tableau Connections 15

- Tableau Connection Properties 16
- Configuring a Tableau Connection 17

Session Configuration for Tableau Targets 18

Appendix A: Data Type Reference 20

Data Type Reference Overview 20

Tableau and Transformation Data Types. 20
Decimal Data Type. 22
Duration Data Type. 22

Index. 23

Preface

Use the *Informatica® PowerExchange® for Tableau User Guide* to learn how to read from or write to Tableau by using PowerCenter Client. Learn to create a Tableau connection, develop mappings, and run sessions in an Informatica domain.

Informatica Resources

Informatica provides you with a range of product resources through the Informatica Network and other online portals. Use the resources to get the most from your Informatica products and solutions and to learn from other Informatica users and subject matter experts.

Informatica Network

The Informatica Network is the gateway to many resources, including the Informatica Knowledge Base and Informatica Global Customer Support. To enter the Informatica Network, visit https://network.informatica.com.

As an Informatica Network member, you have the following options:

- Search the Knowledge Base for product resources.
- View product availability information.
- Create and review your support cases.
- Find your local Informatica User Group Network and collaborate with your peers.

Informatica Knowledge Base

Use the Informatica Knowledge Base to find product resources such as how-to articles, best practices, video tutorials, and answers to frequently asked questions.

To search the Knowledge Base, visit https://search.informatica.com. If you have questions, comments, or ideas about the Knowledge Base, contact the Informatica Knowledge Base team at KB_Feedback@informatica.com.

Informatica Documentation

Use the Informatica Documentation Portal to explore an extensive library of documentation for current and recent product releases. To explore the Documentation Portal, visit https://docs.informatica.com.

If you have questions, comments, or ideas about the product documentation, contact the Informatica Documentation team at infa_documentation@informatica.com.

Informatica Product Availability Matrices

Product Availability Matrices (PAMs) indicate the versions of the operating systems, databases, and types of data sources and targets that a product release supports. You can browse the Informatica PAMs at https://network.informatica.com/community/informatica-network/product-availability-matrices.

Informatica Velocity

Informatica Velocity is a collection of tips and best practices developed by Informatica Professional Services and based on real-world experiences from hundreds of data management projects. Informatica Velocity represents the collective knowledge of Informatica consultants who work with organizations around the world to plan, develop, deploy, and maintain successful data management solutions.

You can find Informatica Velocity resources at http://velocity.informatica.com. If you have questions, comments, or ideas about Informatica Velocity, contact Informatica Professional Services at ips@informatica.com.

Informatica Marketplace

The Informatica Marketplace is a forum where you can find solutions that extend and enhance your Informatica implementations. Leverage any of the hundreds of solutions from Informatica developers and partners on the Marketplace to improve your productivity and speed up time to implementation on your projects. You can find the Informatica Marketplace at https://marketplace.informatica.com.

Informatica Global Customer Support

You can contact a Global Support Center by telephone or through the Informatica Network.

To find your local Informatica Global Customer Support telephone number, visit the Informatica website at the following link:
https://www.informatica.com/services-and-training/customer-success-services/contact-us.html.

To find online support resources on the Informatica Network, visit https://network.informatica.com and select the eSupport option.

CHAPTER 1

Introduction to PowerExchange for Tableau

This chapter includes the following topics:

- PowerExchange for Tableau Overview, 8
- Introduction to Tableau, 8
- PowerExchange for Tableau Implementation, 9
- PowerExchange for Tableau Example, 9

PowerExchange for Tableau Overview

You can use PowerExchange for Tableau to connect to Tableau from PowerCenter.

You can integrate and transform data from sources, such as flat files, databases, and applications to generate a Tableau data extract (TDE) file. You can also create a Tableau packaged workbook (TWBX) and publish the generated file to Tableau.

When you connect to sources directly from Tableau, you have to rely on the speed of the underlying data sources. For faster turnaround, offline access, and to share centralized data with multiple users, you can eliminate connecting to data sources directly from Tableau and use the portable TDE file instead.

The TDE and TWBX files are compatible with Tableau products. You can use the TDE or TWBX file in Tableau Desktop to visualize the data extract and identify patterns and trends. You can also use the Tableau connection in a mapping to publish the TDE or TBWX file directly to Tableau Server or Tableau Online.

Note: PowerExchange for Tableau does not support partitioning.

Introduction to Tableau

Tableau software delivers fast analytics, visualization, and rapid-fire business intelligence.

You can use Tableau Desktop to connect to any data, query the data, see patterns, identify trends, and discover visual insights in seconds. You can create interactive visualizations, reports, and dashboards without the need for programming.

Tableau Server is business intelligence that provides browser-based and mobile analytics. You can publish dashboards to Tableau Server, so that other users can interact with the data in a browser or tablet.

Tableau Online is a hosted version of Tableau Server. You can share dashboards with your organization and customers in minutes. The live, interactive views of data in Tableau Online helps you answer your questions in a web browser or tablet.

PowerExchange for Tableau Implementation

To generate a TDE file from the source data, import the target definitions in the Designer. You can add a target definition to a session and run the session to generate and publish the TDE file to Tableau.

When you specify a Tableau workbook template (TWB) for a Tableau target, the PowerCenter Integration Service applies the TWB template to the TDE file and generates a Tableau packaged workbook (TWBX) file.

The PowerCenter Integration Service integrates with the Tableau data extract API to generate the TDE or TWBX file.

The PowerCenter Integration Service uses the Tableau connection to write the TDE or TWBX file to a directory on the machine where the PowerCenter Integration Service runs. You can publish the TDE or TWBX file to Tableau Server or Tableau Online. The Tableau Rest APIs publish the TDE or TWBX file to Tableau Server or Tableau Online. When you publish the TDE or TWBX file, the file is available for analysis to multiple users within an organization. You can interact with the data, create reports and dashboards from the data, and visually represent the data.

If you do not want to publish the data to Tableau Server or Tableau Online, you can manually import the TDE or TWBX file from the PowerCenter Integration Service machine to Tableau Desktop. You can edit the TDE or TWBX file in Tableau Desktop and later publish the data to Tableau Server or Tableau Online.

PowerExchange for Tableau Example

You are a sales analyst in an enterprise who can access data warehouses or flat files from Tableau Desktop to analyze the data. You want to track the overall growth trend in sales, geographic distribution of sales, and top customers, and present a snapshot of the sales distribution to senior executives.

You can integrate data from multiple sources, filter the data, and make the data available as a TDE file for analysis in Tableau through PowerExchange for Tableau. You can import the TDE file in Tableau Desktop to create interactive, real-time dashboards. The visual representation helps you understand the profitability, with views presented by geography, product category, and customer segment. You can also publish the TDE file to Tableau Server to share a live and interactive dashboard with all the executives in the organization.

CHAPTER 2

PowerExchange for Tableau Configuration

This chapter includes the following topics:

- PowerExchange for Tableau Configuration Overview, 10
- PowerExchange for Tableau Plug-in Registration, 10
- Upgrading PowerExchange for Tableau, 11

PowerExchange for Tableau Configuration Overview

PowerExchange for Tableau installs with PowerCenter.

PowerExchange for Tableau Plug-in Registration

After you complete the installation, register the plug-in with the repository. If you are upgrading from a previous version, update the plug-in registration when you register the plug-in.

To register the plug-in, the repository must be running in exclusive mode. Use the Administrator tool or the pmrep RegisterPlugin command line program to register the plug-in. If you do not have the correct privileges to register the plug-in, contact the user who manages the PowerCenter Repository Service.

The plug-in file is an .xml file that defines the functionality of the adapter. When you install the server component, the installer copies the plug-in file to the following directory: `<PowerCenter installation directory>/server/bin/plugin`

The name of the plug-in file for PowerExchange for Tableau is `tableauPlugin.xml`.

Registering the Plug-in from the Administrator Tool

Register a repository plug-in to add its functionality to the repository.

1. Run the PowerCenter Repository Service in exclusive mode.
2. In the **Navigator**, select the PowerCenter Repository Service to which you want to add the plug-in.
3. In the **Contents** panel, click the **Plug-ins** view.

4. In the **Actions** menu of the **Domain** tab, select **Register Plug-in**.
5. On the **Register Plugin** page, click the **Browse** button to locate the plug-in file.
6. Enter your user name, password, and security domain.

   The **Security Domain** field appears when the Informatica domain contains an LDAP security domain.
7. Click **OK**.

   The PowerCenter Repository Service registers the plug-in with the repository. The results of the registration operation appear in the activity log.
8. Run the PowerCenter Repository Service in normal mode.

Registering the Plug-in from the Command Line Program

You can use the pmrep RegisterPlugin command to register the plug-in from the command line program.

1. Run the PowerCenter Repository Service in exclusive mode.
2. Run the `pmrep Connect` command to connect to the Repository Service using a user account with Administrator Repository privilege.

   The RegisterPlugin command uses the following syntax:

   ```
   pmrep connect -r <repository name> -d <domain_name> -n <domain user name> -x
   <domain_password>
   ```
3. Find `<adaptername>.xml` in the following directory:

   ```
   <Informatica installation directory>\server\bin\Plugin
   ```
4. Run the `pmrep RegisterPlugin` command to update the repository.

   The `RegisterPlugi`n command uses the following syntax:

   ```
   pmrep registerplugin -i <Informatica installation directory>\server\bin\Plugin
   \<adaptername>.xml -e
   ```

Upgrading PowerExchange for Tableau

When you upgrade Informatica services from versions earlier than 10.1, complete the following tasks for PowerExchange for Tableau:

1. Update the plug-in registration when you register the plug-in.
2. You must recreate mappings, if any, if you want to use them in the latest version.

CHAPTER 3

Tableau Targets

This chapter includes the following topics:

- Tableau Targets Overview, 12
- Importing Tableau Target Definitions, 12

Tableau Targets Overview

Tableau target definitions represent metadata based on a Tableau resource.

Use the Target Designer to import Tableau target definitions into the PowerCenter repository. Before you import a Tableau target definition, you need a TDE file to import the metadata.

When you update a TDE file, you can either overwrite the file or append data to the existing file. When you append data to a TDE file, ensure that the column metadata in the TDE file and the Tableau data source are the same. Select the insert option when you edit the Tableau data session properties for Tableau targets so that PowerCenter Integration Service inserts all the rows into the target. You can publish the generated TDE file to Tableau Server or Tableau Online.

Importing Tableau Target Definitions

Import a Tableau target definition from Tableau.

1. In the Target Designer, click **Targets** > **Import from Tableau**.

   The **Import from Tableau** dialog box appears.
2. Enter a valid metadata file directory where you want to import the TDE file.

   Include the full path and the file name. For example, you can specify the following directory:
   `C:\Tableau_Files\TDE_Files\Extract.tde`
3. Click **Connect**, and then click **Next**.
4. From the list of Tableau data extracts, select the record **Extract** to view the fields inside the target.
5. Click **Finish**.

   If you do not specify a file name, the target definition name remains the same as the TDE file name.

CHAPTER 4

Tableau Mappings

This chapter includes the following topics:

- Tableau Mappings Overview, 13
- Tableau Mapping Example , 13

Tableau Mappings Overview

After you import a Tableau target definition into the PowerCenter repository, create a mapping to write data to the Tableau target.

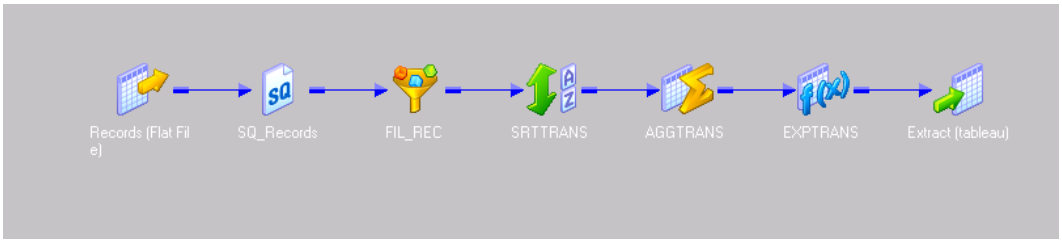
You can read data from multiple sources, write data to the Tableau data extract file, and then publish the file to Tableau Online or Tableau Server.

Tableau Mapping Example

You work in the retail industry, and business analysts in your enterprise need to analyze product sales trends based on region.

Sales record files contain information about products that are sold in multiple outlets and regions. Analysts use flat files to store the sales details. You can consolidate the data in the sales record files that you receive through the day. You can then perform transformations based on your requirements.

The following image shows the Tableau mapping example:



You can use the following objects in a Tableau mapping:

Sources

The mapping contains the Records flat file source that contains the product sales data, such as Region ID, Product ID, Quantity, and Cost.

Transformations

The FIL_REC Filter transformation filters the data in the sales record files based on the value you specify for the region ID. The PowerCenter Integration Service returns the rows that meet the filter condition.

The SRTTRANS Sorter transformation sorts the data in ascending order based on the region ID.

The AGGTRANS Aggregator transformation collects statistics about product sales for a particular region. Use the result of the Sorter transformation as an input to the Aggregator transformation. You can increase Aggregator transformation performance with the sorted input option.

The EXPTRANS Expression transformation formats the data before you generate the Tableau data extract file.

Target

The target extract named tableau is a Tableau data extract file. Select Publish to Server and specify the server details to publish the Tableau data extract file to Tableau Server in the session properties.

When you run the session, the PowerCenter Integration Service writes the sales information to a target TDE file and publishes the TDE file to Tableau Server. You can then visualize the sales data categorized by region in Tableau Server.

CHAPTER 5

Tableau Sessions

This chapter includes the following topics:

- Tableau Session Overview, 15
- PowerExchange for Tableau Connections, 15
- Session Configuration for Tableau Targets, 18

Tableau Session Overview

After you create mappings, you can create a session and use the session in a workflow to extract, transform, and load data. Create sessions and workflows in the Workflow Manager.

To configure the session, perform the following tasks:

- Define the sources from where you consolidate data.
- Configure an application connection for Tableau targets in the Workflow Manager to write data to a TDE file. Provide the Tableau product type to which you want to publish the TDE file.
- Define properties in a session to determine how the PowerCenter Integration Service writes data to a Tableau target TDE file.
- You can apply the Tableau workbook template to the TDE file if you want to create a Tableau packaged workbook (TWBX) file.
- You can configure the session properties for the Tableau target to save the generated TDE or TWBX file to the local machine or to publish the TDE file to Tableau Online or Tableau Server.

PowerExchange for Tableau Connections

Before the PowerCenter Integration Service can connect to Tableau, you must configure a Tableau application connection in the Workflow Manager.

When you configure a Tableau application connection, you specify connection attributes that the PowerCenter Integration Service uses to connect to Tableau. You can specify the Tableau product that you want to connect to.

Tableau Connection Properties

Use a Tableau connection to connect to Tableau. When you create a Tableau connection, you enter information to access Tableau.

The following table describes the Tableau connection properties:

Property	Description
Name	Name of the Tableau connection.
Type	Type of connection. Select Tableau.
Password	Password for the Tableau Server or Tableau Online account.
Connect String	URL of Tableau Server or Tableau Online to which you want to publish the TDE or TWBX file. The URL has the following format: `http://<Host name of Tableau Server or Tableau Online>:<port>`

The following table describes the properties to connect to Tableau:

Connection Property	Description
Tableau Product	The name of the Tableau product to which you want to connect. You can choose one of the following Tableau products to publish the TDE or TWBX file: - Tableau Desktop. Creates a TDE file in the PowerCenter Integration Service machine. You can then manually import the TDE file to Tableau Desktop. - Tableau Server. Publishes the generated TDE or TWBX file to Tableau Server. - Tableau Online. Publishes the generated TDE or TWBX file to Tableau Online.
Connection URL	URL of Tableau Server or Tableau Online to which you want to publish the TDE or TWBX file. The URL has the following format: `http://<Host name of Tableau Server or Tableau Online>:<port>`
User Name	User name of the Tableau Server or Tableau Online account.
Password	Password for the Tableau Server or Tableau Online account.
Content URL	The name of the site on Tableau Server or Tableau Online where you want to publish the TDE or TWBX file. Contact the Tableau administrator to provide the site name.
Template File Path	The path to a sample TDE file from where the PowerCenter Integration Service imports the Tableau metadata. Enter one of the following options for the template file path: - Absolute path to the TDE file. - Directory path for the TDE files. - Empty directory path. The path you specify for the template file becomes the default path for the target TDE file. If you do not specify a file path, the PowerCenter Integration Service uses the following default file path for the target TDE file: `<Informatica Installation Location>\clients\PowerCenterClient\main\java\lib`

Note: You must specify the password of the Tableau Server or Tableau Online account in both the Password fields. You must specify the connection URL of Tableau Server or Tableau Online to which you want to publish the TDE or TWBX file in the **Connect String** field as well.

Content URL

You can specify the name of the content URL to point to a specific site on Tableau Server or Tableau Online where you want to publish the TDE file. Specify the site name in the connection properties.

The content URL has the following format: `http://<Host name of Tableau Server or Tableau Online>:<port> /#/site/<Name of the content URL>/View in Tableau Server or Tableau Online>`

For example, if you create a site called *infa* on Tableau Server, the content URL for the site on Tableau Server is: `https://10.50.100.100:6000/#/site/infa/workbooks`

The value you specify for the content URL in the connection properties is *Infa*.

To specify an existing content URL site name on Tableau Server or Tableau Online where you want to publish the Tableau data extract, contact the Tableau Server or Tableau Online administrator.

Template File Path

When you create a Tableau connection, you can specify the path to a sample TDE file from where the PowerCenter Integration Service imports the Tableau metadata.

The PowerCenter Integration Service generates the target object TDE file from the source object based on the data representation in the specified Tableau template file. The path you specify for the template file in the connection properties becomes the default path for the generated target TDE file.

You can use one of the following options for the template file path:

Absolute path to the TDE file

Enter a directory path along with the TDE file name. For example, enter the following absolute path to a TDE file: `C:\tableau\abc.tde`

Directory path for the TDE files

Enter a directory path that contains the TDE files. For example, enter the following directory path: `C:\tableau`

Empty directory path

If you do not want to use a template file, enter an empty directory path. For example, enter the following directory path: `C:\tableau`

Configuring a Tableau Connection

Before you run a Tableau session, create a Tableau connection.

1. In the Workflow Manager, connect to a repository.
2. Click **Connections** > **Application**.

   The **Application Connection Browser** dialog box appears.
3. Click **New**.

   The **Select Subtype** dialog box appears.
4. To create a Tableau connection, select **Tableau** from the **Select Subtype** list.
5. Click **OK**.

   The **Connection Object Definition** dialog box appears.
6. Enter the connection properties.
7. Click **OK**.

   The Tableau connection appears in the **Connection Browser** list.

Session Configuration for Tableau Targets

You can configure the session properties for a Tableau target in the **Transformations** view on the **Mapping** tab. Define the properties for the target instance in the session.

The following table describes the session properties that you can configure for Tableau targets:

Target Operation

Creates, overwrites, or appends the TDE file on the local machine, Tableau Server, or Tableau Online.

Select one of the following options to publish the TDE file:

- Create. Creates a TDE file. Ensure that a TDE file with the same name does not exist.
- Append. Adds data to an existing TDE file.
- Overwrite. Deletes the existing TDE file and creates a new TDE file.

Note: The append operation works only if there is an existing TDE file.

Extract File Path

The file path where you want to save the generated Tableau data extract file.

Default is `<INFA_HOME>/server/bin`. Ensure that the file location is on the machine where the PowerCenter Integration Service runs. You require the write permissions on the `<INFA_HOME>/server/bin` file location.

If you have specified the template file path in the Tableau connection properties, but not an extract file path, the PowerCenter Integration Service considers the template file path as the extract file path.

Extract File Name

Name of the TDE file with the .tde extension. Default is `Extract.tde`.

If the operation is for Tableau Server or Tableau Online, the PowerCenter Integration Service deletes the file after publishing the TDE file to Tableau Server or Tableau Online.

Project Name

Name of the project within a specific site on Tableau Server or Tableau Online where you want to publish the TDE file. By default, Tableau Connector publishes the TDE file to the default project on the site that you specify.

Data Source

Name of the Tableau data extract that you want to publish to Tableau Server or Tableau Online. If you do not specify a data source name, the default Tableau data extract file name remains the source name.

Workbook Template File name

Name of the predefined Tableau workbook template (TWB) file name that you want to apply to the TDE file to generate a Tableau packaged workbook file (TWBX). You must provide the TWB name if you want to publish the TWBX file.

Workbook Name

Name for the workbook that you want to publish to Tableau. If you do not specify a workbook name, the name of the TWB template file remains the workbook name.

INSERT

Inserts all the rows to the target TDE file. You must select the INSERT option before you run a session.

DELETE

Not applicable.

UPDATE

Not applicable.

Success File Directory

Reserved for future use.

Error File Directory

Reserved for future use.

APPENDIX A

Data Type Reference

This appendix includes the following topics:

- Data Type Reference Overview, 20
- Tableau and Transformation Data Types, 20
- Decimal Data Type, 22
- Duration Data Type, 22

Data Type Reference Overview

PowerCenter uses the following data types in Tableau mappings:

- Tableau native data types. Tableau data types appear in the Tableau definitions in a mapping.
- Transformation data types. Set of data types that appear in the transformations. They are internal data types based on ANSI SQL-92 generic data types, which the PowerCenter Integration Service uses to move data across platforms. Transformation data types appear in all transformations in a mapping.

When the PowerCenter Integration Service reads source data, it converts the native data types to the comparable transformation data types before transforming the data. When the PowerCenter Integration Service writes to a target, it converts the transformation data types to the comparable native data types.

Tableau and Transformation Data Types

The following table lists the Tableau data types that the PowerCenter Integration Service supports and the corresponding transformation data types:

Tableau Data Type	Transformation Data Type	Range and Description
Integer	Integer	-2,147,483,648 to 2,147,483,647 Precision 10, scale 0
Double	Double	Double-precision floating-point numeric value. Precision 15

Tableau Data Type	Transformation Data Type	Range and Description
Date	Date/Time	Jan 1, 0001 A.D. to Dec 31, 9999 A.D. Precision of 29, scale of 9 (precision to nanosecond)
DateTime	Date/Time	Jan 1, 0001 A.D. to Dec 31, 9999 A.D. Precision of 29, scale of 9 (precision to nanosecond)
unicode_string	String, Text, Bigint, or Decimal	The PowerCenter Integration Service performs an implicit conversion of String, Text, Bigint, or Decimal to unicode_string: String: - 1 to 104,857,600 characters - Fixed-length or varying-length string Text: - 1 to 104,857,600 characters - Fixed-length or varying-length string Bigint: - 9,223,372,036,854,775,808 to 9,223,372,036,854,775,807 - Precision of 19, scale of 0 - Integer value Decimal: - Precision 1 to 28 digits, scale 0 to 28 - Decimal value with declared precision and scale. Scale must be less than or equal to precision.
char_string	String, Text, Bigint, or Decimal	The PowerCenter Integration Service performs an implicit conversion of String, Text, Bigint, or Decimal to char_string: String: - 1 to 104,857,600 characters - Fixed-length or varying-length string Text: - 1 to 104,857,600 characters - Fixed-length or varying-length string Bigint: - 9,223,372,036,854,775,808 to 9,223,372,036,854,775,807 - Precision of 19, scale of 0 - Integer value Decimal: - Precision 1 to 28 digits, scale 0 to 28 - Decimal value with declared precision and scale. Scale must be less than or equal to precision.
boolean	String	1 to 104,857,600 characters. Fixed-length or varying-length string. Valid values are True and False.
duration	String	Valid values for hours are integer values between 0 and 23. Valid values for minutes and seconds are integer values between 0 and 59. If there is no value for any field, specify 0.

Decimal Data Type

When you read data as Decimal in the source definition, use String or Double instead of the Decimal data type for better performance. As Tableau does not support the Decimal data type, you must set the decimal data to string data types that Tableau supports. Change the decimal data type to string in the source definition and char_string or unicode_string in the target definition. The PowerCenter Integration Service performs an implicit conversion of decimal to a comparable native data type, unicode or char_string, that Tableau supports.

Duration Data Type

Duration is specified in days, hours, minutes, seconds, and milliseconds. All the values must be integers. You must change the string data type that arrives from different source fields to a single string value and then map this string value to the duration data type in the target.

For example, the PowerCenter Integration Service reads data from five different source fields of string data type, such as, 5 days, 10 hours, 21 minutes, and 35 seconds. Use the Expression transformation to concatenate the input string values to a single string value of comma-separated values, such as 5,10,21,35,0. In the target definition, map the string output received from the Expression transformation to duration data type. Use the single string value as the input value and duration as the output value in the target.

INDEX

C

connection properties
  content URL 17
  template file path 17
content URL
  connection properties 17

D

data type reference
  overview 20

P

plug-ins
  registering 10
PowerExchange for Tableau
  configuration overview 10
  data types 20
  overview 8
  sessions overview 15
PowerExchange for Tableau connections
  configuring 17

R

registering
  plug-ins 10
registering plug-ins
  PowerExchange for Tableau 10

S

session properties
  Tableau targets 18

T

Tableau
  targets overview 12
Tableau mappings
  example 13
Tableau session properties
  configuration 18
Tableau targets
  importing 12
template file path
  connection properties 17

U

upgrading
  PowerExchange for Tableau upgrade tasks 11